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Key Points:

- Hillslope connectivity shows high spatial variability across the United States
- Tectonics is a primary control of the strength of structural hillslope connectivity
- Open access data portal of connectivity maps created for stakeholders

Supporting Information:

Supporting Information may be found in the online version of this article.

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Structural Hillslope Connectivity Is Driven by Tectonics More Than Climate and Modulates Hydrologic Extremes and Benefits

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Abstract Structural hydrologic connections, formed by the coaction of tectonic and climatic processes, facilitate the transfer of matter across Earth's surface. Large-scale analysis of hillslope connectivity controls is lacking, thus limiting our understanding of the drivers of extremes, like floods and landslides, and benefits, like wetlands. Here, we model hillslope connectivity using the Index of Connectivity, at 10-m resolution for the contiguous United States. We show the dominance of tectonic drivers, like river steepness ($\rho = 0.84$) and seismic activity ($\rho = 0.48$), over climatic drivers, like precipitation ($\rho = -0.31$) and aridity ($\rho = -0.07$), in controlling the strength of connectivity. Highly connected basins were associated with landslide occurrence ($\rho = 0.69$) while poorly connected basins had greater wetland density ($\rho = -0.50$). Lastly, we share our results in an open access data portal for land managers to leverage structural connectivity maps in their respective study domains.

Plain Language Summary Hillslopes are critical landscape features that intercept, store, and route water, from its source as rainfall to its fate as river discharge. The strength of this routing is a function of climatic and tectonic forces, but their relative importance to hillslope connectivity is uncertain. We simulated the Index of Connectivity, a topographic analogue for structural connectivity, for 75 billion locations in contiguous United States (CONUS), across a range of climatic and tectonic settings. At the CONUS-scale, we found that hillslope connectivity is largely driven by tectonic forces, including uplift and seismic activity, and that highly connected hillslopes are more susceptible to landslides while poorly connected hillslopes promote wetland development. We provide a web data portal to serve as a tool for stakeholders to visualize and leverage structural connectivity data in their respective study areas.

1. Introduction

Hillslopes are dynamic landscape features that connect ridges to valleys, linking terrestrial and aquatic ecosystems through water, sediment, and nutrient fluxes (Fan et al., 2019; Stieglitz et al., 2003). The degree to which a hillslope parcel is connected to a downstream waterbody is a function of the energy inputs into the parcel and the likelihood for disconnectivity during downstream transport (Najafi et al., 2021). The strength of this connection is a product of tectonic and climatic drivers (Montgomery & Dietrich, 1992; Whittaker, 2012). Recent large-scale analyses of rivers, which act as end-points to the hillslope continuum, disagree whether tectonic or climatic processes control longitudinal river structure (Chen et al., 2019; Seybold et al., 2021). However, similar large-scale analyses of hillslope connectivity are lacking despite being crucial for understanding how landscapes adapt to climate change and other anthropogenic stressors.

Structural (or “static”) connectivity sets the long-term propensity for downgradient transport whereas functional (or “dynamic”) connectivity varies over the short-term and includes the interplay of spatial and temporal energy fluxes (Bracken et al., 2015; Mahoney et al., 2020a, 2020b). For example, short-term meteorological conditions may characterize the functional hillslope response whereas long-term climatic conditions may set structural landscape evolution trends. To this end, structural connectivity is informed by landscape topographic properties, including elevation, slope, aspect, curvature, and roughness (Heckmann et al., 2018). Further, topography serves as a “tape recorder” of the history of a landscape as tectonic and climatic drivers leave behind features through uplift and downcutting that encode their historic influence on connectivity (Whittaker, 2012).

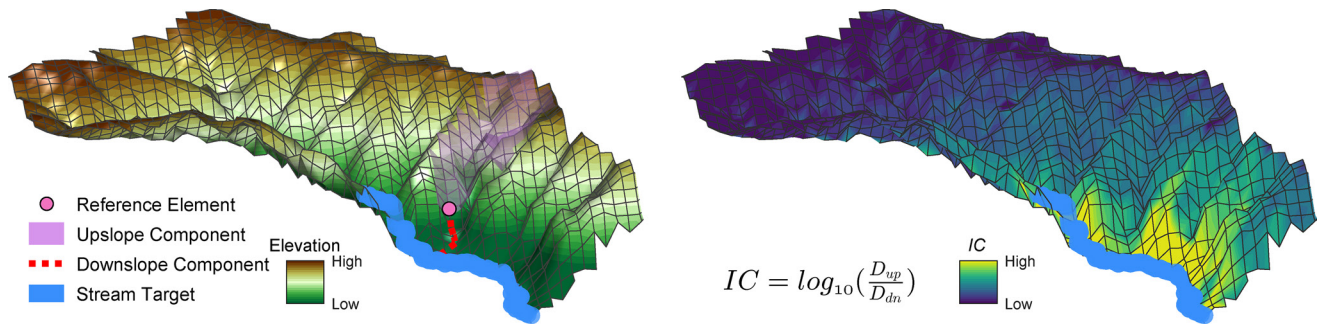


Figure 1. Conceptual diagram of Index of Connectivity (IC) calculation, which utilizes weighted upstream (D_{up}) and downstream (D_{dn}) components. IC , D_{up} , and D_{dn} are defined in Equations 2–4 in Section 2.

Geomorphometric indices, derived from digital elevation models (DEM), show utility for inferring hydrologic and geomorphologic conditions from topographic data (Heckmann et al., 2018; Najafi et al., 2021). One such tool, the Index of Connectivity (IC) is a dimensionless parameter that indicates the potential strength of linkage between a hillslope parcel and a stream target (Figure 1; Borselli et al., 2008). IC has been applied with success to individual hillslopes (Heckmann & Vericat, 2018), urbanizing basins (Michalek et al., 2021), lowland environments (Gay et al., 2016), and alpine terrain (Cavalli et al., 2013). However, IC is typically computed in one-off instances and its potential to explain hydrologic drivers of connectivity across large spatial domains has not been evaluated.

Drivers of hydrologic connectivity broadly include tectonics and climate. Tectonic drivers, such as river structure (steepness and concavity) and seismic activity, can reveal long-term uplift and downcutting dynamics (Seybold et al., 2021). On the other hand, climatic drivers, such as precipitation and aridity, can indicate the strength of hillslope-channel connectivity. While recent observations have suggested that arid (Chen et al., 2019) and wet (Harries et al., 2021) climates produce low and high connectivity, respectively, other work has found that step-changes in climate produce unexpected, non-linear responses (Leonard & Whipple, 2021). High connectivity may allow for efficient drainage of landscapes, leading to increased flash-flooding hazards (Hallema et al., 2016), and it can also provide the necessary energy for large mass-failure events (Schildgen et al., 2016). Connectivity can also confer landscape benefits, such as in the case of low connectivity providing long residence times necessary for wetland development and effective bioremediation of contaminants (Cohen et al., 2016; Golden et al., 2014).

The objective of this study was to assess tectonic and climatic controls on structural hillslope connectivity and to infer how connectivity may impact the occurrence of floods, landslides, and wetlands. We use three resolutions of DEM data (90-m, 30-m, and 10-m) to model IC at a total of 75 billion hillslope points across contiguous United States (CONUS). To assess connectivity controls, we compare IC results to geomorphometric, earthquake, and land-surface modeling predictions. To infer hydrologic extremes and benefits, we relate the strength of basin-scale IC to the occurrence of rainfall-induced landslides, river-flooding events, and wetland density estimates. Lastly, we develop a web data portal to serve as a tool for stakeholders to visualize and leverage structural connectivity data in their respective study areas.

2. Materials and Methods

2.1. Hillslope and River Geomorphometrics

To calculate geomorphometric indices of interest (k_s , θ , and IC), we first retrieved the necessary DEM products from the United States Geologic Survey (USGS, 2019) (Figures S1–S4 in Supporting Information S1). Thereafter, all calculations were performed for hydrologic unit code level six (HUC-6) basins, of which there are 332 in CONUS with an average area of 25,000 km² (see Supplemental Data in Supporting Information S1 for basin details).

The first two indices can be retrieved from a power law relationship (Hack, 1957), which states that channel slope (S) systematically declines with increasing drainage area (A):

$$S = k_s A^{-\theta}, \quad (1)$$

where k_s is the steepness index, which describes the overall gradient of the channel, and θ is the concavity index, which describes how rapidly the channel gradient decreases downstream. k_s and θ values were calculated on the trunk channel for each basin using *TopoToolbox* (Schwanghart & Scherler, 2014).

The last index, IC , is defined as (Borselli et al., 2008):

$$IC = \log_{10} \left(\frac{D_{up}}{D_{dn}} \right), \quad (2)$$

where D_{up} and D_{dn} represent the upslope and downslope elements of connectivity, respectively (Figure 1). IC can range from $[-\infty, +\infty]$ with larger values indicating greater structural connectivity. The upslope component D_{up} is the potential for down routing of hillslope sediment:

$$D_{up} = \overline{W} \overline{S} \sqrt{A}, \quad (3)$$

where $\overline{W}$ is the average upslope weighting factor, $\overline{S}$ is the average upslope gradient, and A is the contributing upslope area. The downslope component D_{dn} is the sinking potential along the downslope flow path with the path being discretized into i -increments:

$$D_{dn} = \sum \frac{d_i}{W_i S_i} \quad (4)$$

where d_i is the downslope increment length along the steepest slope direction, W_i is the downslope weighting factor, and S_i is the slope gradient at the i^{th} increment. For space considerations, further geomorphometric procedure details are expounded in the Supplementary Methods in Supporting information S1.

2.2. Tectonic and Climate Drivers Data

To assess tectonic drivers, we assimilated the results of the Global Earthquake Model (GEM) to estimate seismic activity across CONUS (Pagani et al., 2020). GEM depicts, at 4-km resolution, the peak ground acceleration (PGA) that has a 10% probability of exceedance in 475 years based on earthquake observations and seismic wave propagation models. To assess climatic drivers, we utilized the primary forcing variables from the North American Land Data Assimilation System at 14-km resolution (Mitchell et al., 2004), including precipitation and evapotranspiration. The aridity index was computed as the ratio of precipitation to potential evapotranspiration, that is, $AI = P/PET$ (Chen et al., 2019). For each HUC-6 basin, mean driver values were determined by averaging over the areal basin extent. To generate HUC-2 region results, all basin values contained within a region were area-weighted and averaged together.

2.3. Hydrologic Extremes and Benefits Data

Rainfall-triggered landslide events were retrieved from the Global Landslide Catalog (Kirschbaum et al., 2010), which documents events from news reports, scholarly articles, and other hazard databases. In total, 3,502 CONUS landslide events were recorded in the catalog, occurring between 1996 and 2017. The Dartmouth Flood Observatory (DFO) database was used to retrieve the location of significant river-flooding events, which are compiled from news, government agencies, stream gages, and remote sensing (Li et al., 2021; Tellman et al., 2021). In total, from 1985 to 2020, 424 CONUS river-flooding events were documented in the database. Wetland coverage was estimated from the 2019 National Land Cover Database (MRLC, 2019).

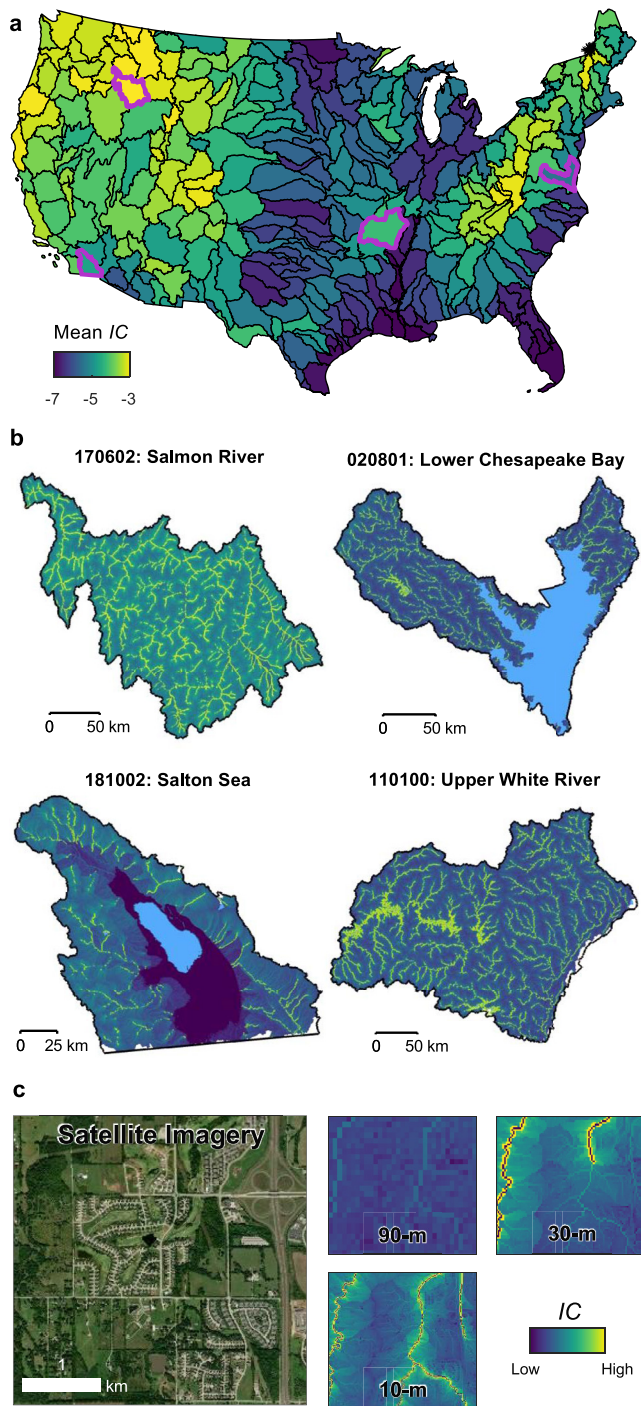


Figure 2. (a) Mean Index of Connectivity (IC) for all hydrologic unit code level six (HUC-6) basins. (b) Four basins highlighted in (a) are expanded to demonstrate spatial variability in IC . (c) Comparison of IC derived from three digital elevation model (DEM) resolutions for a 9 km² field site in eastern Kansas (HUC-6: 102701).

3. Results and Discussion

3.1. Continental Gradients in Hillslope Connectivity Across Scales

Structural hillslope connectivity varies considerably across CONUS with the largest basin-averaged IC values in the Rocky Mountains, the Appalachian Mountains, and the Pacific Northwest while the lowest values appear near the Gulf of Mexico, the southern Atlantic coast, and the Great Plains (Figure 2a). While basin-averaged IC is a general indicator of connectivity, it is important to note that even in basins with low overall IC , there can exist zones of elevated connectivity. To demonstrate this variability, we highlight four such HUC-6 basins: Salmon River, Lower Chesapeake Bay, Salton Sea, and Upper White River (Figure 2b). The Salmon River basin ranks as one of the most highly-connected in our analysis with steep, glaciated slopes and erosive soils throughout the basin. On the other hand, basins with large waterbodies, such as the Lower Chesapeake Bay and the Salton Sea, are characterized by higher connectivity close to stream reaches, but lower connectivity along ridges and locations distal to hydrologic sinks. Inland reservoirs, like those in the western and central regions of the Upper White River, which were constructed for hydroelectric power generation and flood control, increase the lateral connectivity of surrounding hillslopes to stream networks by expanding the potential sink area for mobilized water and sediment. However, while reservoirs may increase lateral connectivity of sediment feeding into them, they dramatically decrease longitudinal downstream connectivity by forcing siltation (Kondolf et al., 2014).

Microtopographic features can also serve as connectivity hot spots, which we demonstrate with a comparison of IC derived from three DEM resolutions (90-m, 30-m, and 10-m) for a 9 km² field site in eastern Kansas (Figure 2c). At the coarsest resolution, only the major stream channels are identified as having high connectivity. As resolutions become finer, small tributaries are identified as high connectivity while ridges are identified as low connectivity. Finally, at the 10-m resolution, high connectivity features such as road-side ditches and the banks of a cloverleaf traffic interchange are identified. On average, high connectivity hot spots make up 0%–5% of a basin area, with higher resolution data revealing twice the fraction of hot spots. In reality, this percentage may be even higher as our model doesn't capture manmade subsurface drainages, like tile and storm drains, which are not represented in surficial DEM data, but may impact connectivity (Michalek et al., 2021). Further, in our analysis, we weigh IC components (D_{up} and D_{dn}) using microtopographic undulations whereas other studies have considered the RUSLE C-factor and the Manning's roughness coefficient as weights (Hooke et al., 2021). Smaller-scale analysis should carefully select the appropriate weighting scheme for the particular application.

Close inspection of the modality and statistical properties of structural connectivity for all 332 HUC-6 basins revealed three prevailing archetypes (Figure 3): (a) low-connectivity basins with high kurtosis and low variance, (b) intermediate-connectivity basins with low kurtosis and high variance, and (c) high-connectivity basins with low kurtosis and low variance. In Archetype 1 (low-connectivity), upstream energy inputs are small relative to downstream distance to a sink and IC values tightly center around a low overall value.

Examples include most basins in the South Atlantic-Gulf and Lower Mississippi HUC-2 regions (Figure S5 in Supporting Information S1). For Archetype 2 (intermediate-connectivity), basins comprise areas of both high and low connectivity and may have varied geomorphology, such as a combination of plains and mountains. Examples include the Rio Grande Headwaters (130100) and many basins in the

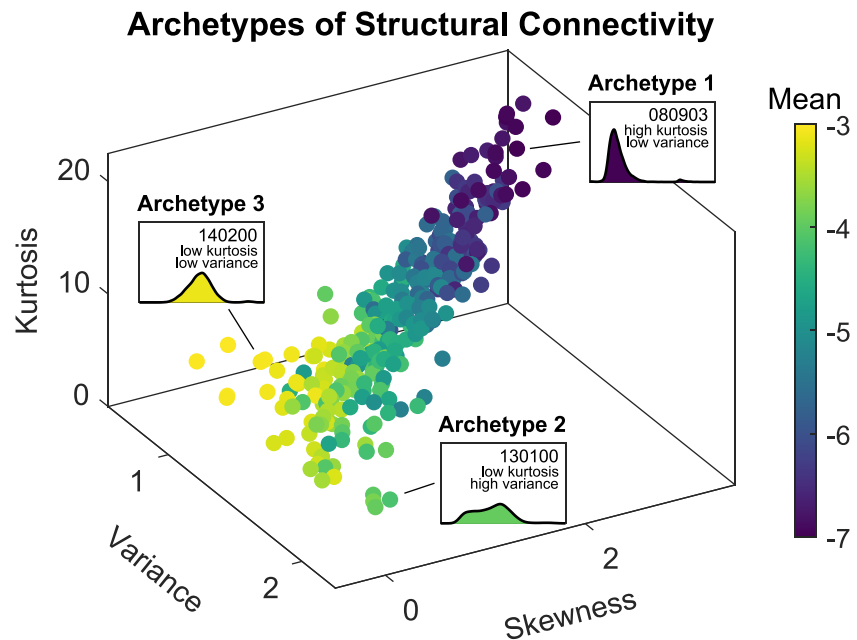


Figure 3. The statistical moments of basin-scale Index of Connectivity (*IC*), including mean, variance, skewness, and kurtosis. Three archetypes are revealed. Inset images are labeled with hydrologic unit code level six (HUC-6) basin identifier and show *IC* density functions as in Figure S5 in Supporting information S1.

Great Basin and California regions, including the Jordan River (160202) and San Joaquin River (180400). Lastly, in Archetype 3 (high-connectivity), upstream energy inputs are large and high *IC* is generally observed though not as centrally clustered about a single point as in Archetype 1. Examples include the Gunnison Basin (140200) and about one-third of the Pacific Northwest region where steep mountainous terrain provides surficial energy inputs.

3.2. Climatic and Tectonic Drivers of Connectivity

Both tectonic and climatic drivers exhibit spatial variability across the CONUS landmass with the potential to explain variabilities in structural connectivity (Figure S6 in Supporting Information S1). Strong spatial coherence exists between structural connectivity and all tectonic drivers: PGA, river steepness, and channel profile concavity (Figures 4a–4c). At the basin-level, the correlations of *IC* with river steepness ($\rho = 0.84$) and concavity ($\rho = 0.52$) are high, confirming that when the primary river channel is steep and highly concave, structural hillslope connectivity to the river network is high. Likewise, *IC* correlates strongly with PGA ($\rho = 0.48$) from the GEM (Pagani et al., 2020), suggesting that historical seismic activity amplifies the structural connectivity of a landscape.

Weak spatial coherence exists between *IC* and all climatic drivers: precipitation, potential evapotranspiration, and aridity index (Figures 4d–4f). Slight negative correlations of basin-average *IC* and precipitation ($\rho = -0.31$) and potential evapotranspiration ($\rho = -0.35$) suggest wetter basins have lower structural connectivity. However, it is important to note that these totals are evaluated annually and the drivers shaping functional connectivity—which differs from structural connectivity—may occur at shorter time-scales, such as in 24-hr rainfall intensity (Sweeney et al., 2015; Zaprowski et al., 2005). Notably, the relationship between *IC* and the aridity index is very weak ($\rho = -0.07$), indicating that structural hillslope connectivity is not influenced by aridity. Regarding river channels, some authors suggest aridity is a first-order control (Chen et al., 2019) whereas others disagree in favor of tectonics (Seybold et al., 2021). To extend this discussion to hillslopes, which are tightly coupled to—and terminate at—river networks, our results suggest that aridity does not play a significant role in controlling structural hillslope connectivity. Rather, our results indicate that tectonics is a first-order control, and that climate is of lesser influence. Importantly, while spatial correlation between structural connectivity and the enumerated drivers may exist, this does not directly imply a causal relationship. While we have confidence in our findings,

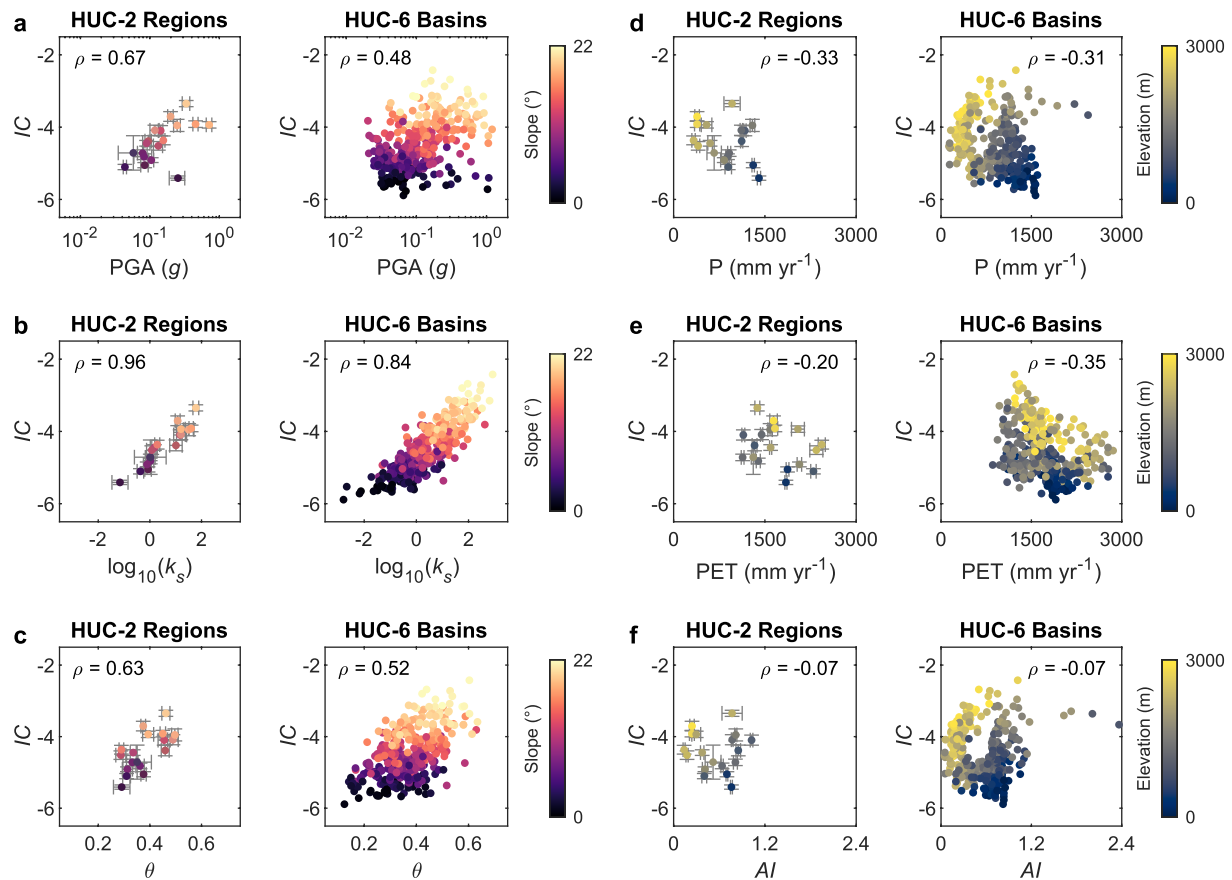


Figure 4. Correlations between Index of Connectivity (IC) and measures of tectonics and climate at the hydrologic unit code level two (HUC-2) region-scale ($n = 18$) and the HUC-6 basin-scale ($n = 332$). Tectonic measures include (a) peak ground acceleration, (b) the steepness index, and (c) the concavity index. Climatic measures include (d) precipitation, (e) potential evapotranspiration, and (f) aridity index. The Spearman rank correlation (ρ) is shown and the error bars display the standard error for each HUC-2 region as calculated from all basins within a region. Tectonic measures are color-coded by mean watershed slope whereas climatic measures are color-coded by mean elevation.

there exists potential to further reduce uncertainty through additional modeling in geomorphological settings that may experience different climatic and tectonic drivers.

The mechanism of tectonic control on structural hillslope connectivity manifests through the spatial variability in uplift rates (Burbank & Anderson, 2011). Areas with high tectonic activity, that is, those with substantial rifting and subduction, experience long-term spatial contrasts in uplift, which creates a landscape characterized by steep topography (Seybold et al., 2021). In our conceptualization of structural hillslope connectivity (Figure 1), slope is both a major influence of upstream energy inputs and downstream pathing of material. Thus, in basins where tectonic influences are prevalent, slopes are steep, and energy for transport is available, hillslopes are able to effectively retrieve and transfer flow and sediment inputs from ridges to valley bottoms (Sweeney et al., 2015).

3.3. Hydrologic Extremes, Benefits, and Management

The occurrence of hydrologic extremes (landslides and floods) and benefits (wetlands) shows high spatial variability across CONUS (Figure 5). Basin-scale structural connectivity was strongly correlated with the occurrence of rainfall-induced landslides ($\rho = 0.69$), highlighting the utility of IC as a predictive tool for some hydrologic extremes. However, the relationship between structural connectivity and large river-flooding events was very weak ($\rho = -0.12$), indicating that certain hydrodynamic processes may not be well-represented by static landscape variables alone. Lastly, wetland coverage was inversely related to connectivity ($\rho = -0.50$), corroborating studies demonstrating that poor structural connectivity has a role in wetland function (Cohen et al., 2016; Golden et al., 2014).

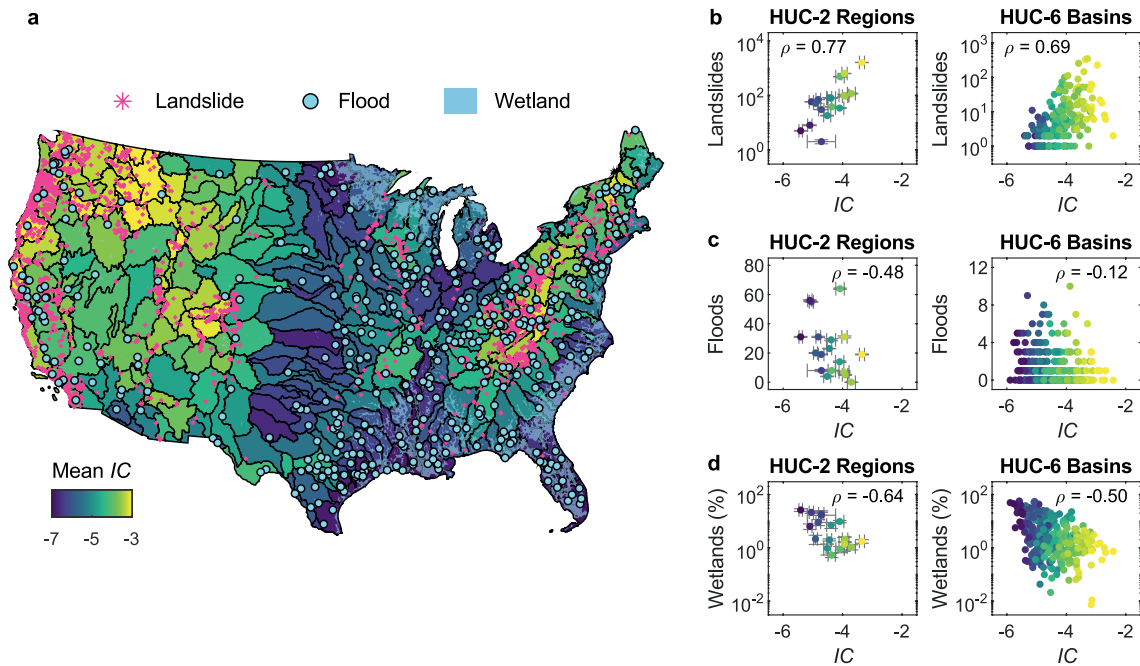


Figure 5. (a) Spatial distribution and correlations of Index of Connectivity (IC) with the occurrence hydrologic extremes and benefits, including (b) landslides, (c) floods, and (d) wetlands. The Spearman rank correlation (ρ) is shown, and the error bars display the standard error for each hydrologic unit code level two (HUC-2) region as calculated from all basins within a region. Scatter points are color-coded by mean watershed slope from 0° (blue) to 22° (yellow).

We originally hypothesized that flooding would occur primarily in low structural connectivity basins because of poor water drainage and limited conveyance capacity of low-sloping hillsides. However, this relationship largely did not bear out. This could be for a few reasons, including (a) floods may be strongly impacted by local heterogeneities that are not reflected in basin-scale mean statistics and (b) floods in remote or uninhabited areas, which also happen to have the highest IC , may be under-sampled since DFO flood records ($n = 424$) generally align with population density. To this last point, nearly half of the basins in our study had no recorded floods in the CONUS DFO database; more data could help to alleviate these issues and develop clearer linkages between data. For example, the global DFO database has 4,968 total events, an order of magnitude more than CONUS events, and future analysis at this scale could resolve uncertainties.

A limitation of our overall modeling approach is that we exclusively consider static surface characteristics. In reality, subsurface processes, such as water storage and transmissivity, can be of considerable importance and can even lead to different hillslope structure given similar tectonic and climatic forcings (Litwin et al., 2021). A more explicit modeling of the dynamic fluxes between the surface, subsurface, and deeper groundwater could potentially improve prediction of hydrologic extremes. However, this kind of dynamic connectivity model requires considerable data and modeling resources, which are underdeveloped at continental scales. We suggest building upon existing connectivity models, for example, CASCADE (Catchment Sediment Connectivity And Delivery model) (Hooke & Souza, 2021; Schmitt et al., 2016; Tangi et al., 2019), to expand into new spatiotemporal scales and geomorphological domains. Advancements in high-performance computing and deep learning algorithms are making these kinds of efforts more viable than ever before (Razavi, 2021; Sit et al., 2020).

While scientific advancements are crucial to furthering our understanding of landscape connectivity, ultimately these results must be translated into useable tools for land managers. Recent work has highlighted that over three-quarters of stakeholders were unaware of recent developments in connectivity research (Smetanová et al., 2018). Half of these stakeholders attributed the disconnect, at least in part, to a lack of data availability and practical methodology (Smetanová et al., 2018). This is a missed opportunity for scientists as nearly all stakeholders who were aware of connectivity-related linkages used that knowledge to influence their management (Smetanová et al., 2018). We hope that by making our basin-scale IC maps easy-to-access, they can serve as a useful tool for stakeholders interested in visualizing structural connectivity. Our connectivity maps can be freely

and openly accessed through the Consortium of Universities for the Advancement of Hydrologic Science Inc. (CUAHSI) HydroShare web-tool: <https://apps.cuahsi.org/connectivity-map>.

4. Conclusion

In this study, we calculated *IC* at 75 billion points to explain the spatial variability of structural hillslope connectivity. We demonstrated the predominance of tectonics over climate for explaining *IC* across CONUS basins with river steepness ($\rho = 0.84$) as the most prominent driver. Further, we related the magnitude of *IC* to the occurrence of landslides, floods, and wetlands, and we found that landslides and *IC* have a strong positive correlation ($\rho = 0.69$) while wetlands and *IC* have a strong negative correlation ($\rho = -0.50$). We provide a better understanding of the drivers and consequences of connectivity as anthropogenic forces shape climate and geomorphology into the future.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

All data and code generated in this study are available in the Supporting Information S1 or at CUAHSI HydroShare: <https://doi.org/10.4211/hs.e80005f70e974e729db02b25945ded8d>. The earthquake, landslide, and flood data generated by other researchers and used in this study are available in Pagani et al. (2020), Kirschbaum et al. (2010), and Li et al. (2021), respectively.

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